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Optimization and assessment of carbon capture, transport and storage supply chains for industrial sectors: The cost of resilience

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ABSTRACT

This study investigates the optimal design of carbon capture, transport and storage (CCTS) supply chains for decarbonizing industrial emitters. It builds upon previous analyses, which determine the cost-optimal design of CCTS supply chains while complying with specified emissions reduction scenarios, and determines optimal designs in terms of minimum cost and maximum resilience. The resilience of a supply chain is defined as its ability to permanently store the captured CO_2 during the time horizon of interest, and it is quantified in terms of expected amount of carbon not stored under a fixed number of possible simultaneous failures. A mixed-integer linear program is developed to design CCTS supply chains that ensure a specified level of resilience while minimizing total system costs and CO_2 emissions. The model is illustrated by determining the optimal decarbonization strategy in terms of costs and level of resilience for the Swiss waste-to-energy sector from 2025 to 2050. Overall, resilience is achieved with a cost increase with respect to the cost-optimal solution ranging from about 5% (backup truck connections to minimize the cost of the supply chain) to about 70% (backup pipeline connections to minimize the emissions of the supply chain) when the storage site is in the North Sea. When the storage site is available in Switzerland, the cost increase is much reduced and ranges from about 1% to 10% with respect to the cost-optimal solution.

1. Introduction and scope

1.1. Context: Relevance of carbon capture, transport and storage (CCTS) infrastructure

In light of the reports released by the Intergovernmental Panel on Climate Change (IPCC), new pathways must be explored to reach the net-zero-CO_2-emissions goal by 2050 (Intergovernmental Panel on Climate Change, 2018; IPCC, 2021). One component of such pathways consists of carbon capture and storage (CCS), which is especially relevant for reducing the emissions of hard-to-decarbonize industrial sectors, such as the chemical (Gabrielli et al., 2020b), the aviation (Becattini et al., 2021), or the cement industries (Benhelal et al., 2013; Fennell et al., 2021). Whereas CCS hubs and clusters have been developed recently to aggregate CO_2 streams captured at different facilities (Equinor, 2019; Porthos Project, 2021; Carbfix, 2021; Ringrose, 2020), one of the major barriers for the wide deployment of CCS is given by the design, development and deployment of a global CO_2 network connecting sources and sinks across different countries (Buit et al., 2011; Morbee et al., 2010). In other words, more attention should be paid to the transport step and to the analysis of carbon capture, transport and storage (CCTS) supply chains, which are defined as a network of CO_2 emissions and storage facilities that can be connected by means of various transport options.

A number of studies have proposed decision-support tools for the design and operation of CCTS supply chains. Most studies focus on the design of cost-optimal supply chains that meet specified environmental goals (Bakken and Velken, 2008; Morbee et al., 2012; Oei et al., 2014; Elahi et al., 2014; Jakobsen et al., 2014; d'Amore and Bezzo, 2017; Middleton et al., 2020; Wei et al., 2021). In an earlier work, we build upon such studies by formulating and solving an optimization problem, based on mixed integer linear programming (MILP), to design minimum-cost CCTS supply chains that comply with specified time-dependent CO_2 emissions reduction scenarios. This model is applied to determine the optimal infrastructure rollout to decarbonize the Swiss waste-to-energy (WtE) sector from 2025 to 2050 (Becattini et al., 2022), which is especially relevant due to the possibility of generating net-negative CO_2 emissions (Lausselet et al., 2017; Fuss et al., 2014; Becattini et al., 2022). We now expand the analysis to determine the optimal design of CCTS supply chains in terms of minimum costs and maximum resilience, focusing specifically on CO_2 transport.

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Within the CCS domain, the notion of risk has been explored from different angles and with a variety of techniques. Zhang et al. investigated the effect of uncertain policy and technology performance on the financial risk associated with the supply chains (Zhang et al., 2018, 2019), whereas Bjerketvedt et al. studied the use of buffer storage to increase reliability under uncertainty in CO_2 ship-based transport (Bjerketvedt et al., 2020). Approaches have been suggested to integrate the risk associated with uncertain geological storage capacities in the design of CCTS supply chains (d'Amore et al., 2019; Lane et al., 2021), to understand the chain of events that could lead to failures making use of fault tree analysis (Oraee-Mirzamani et al., 2013), and to consider and mitigate societal risks (D'Amore et al., 2018). While not focusing on CCS, Al-Yaeeshi and Al-Ansari developed a framework to assess and maximize the operational resilience of CO_2 utilization networks (Al-Yaeeshi and Al-Ansari, 2022). However, resilience to failures and its impact on the design and costs of CO_2 networks for CCTS supply chains have not been defined and investigated so far.

1.2. Scope: Cost-optimal and resilient CO_2 transport networks

Information on the reliability and resilience of a system, through concepts such as the N-1 contingency criterion, is highly valuable not only for infrastructure planning but also for its management (Hadjidimitriou et al., 2021). Therefore, the reliability and resilience of CCTS supply chains, encompassing the capture, transport, and storage stages, need to be considered already at design phase. This contribution defines the concept of resilience for CCTS supply chains, and it develops an optimization model for the design of cost-optimal and resilient CO_2 networks, i.e. focusing on resilience to failures in CO_2 transport.

The model builds upon a MILP optimization framework presented earlier (Gabrielli et al., 2018a, 2020a; Becattini et al., 2022). The original model minimizes the costs (investment, operation and maintenance) of CCTS supply chains while complying with CO_2 emissions reduction scenarios (Becattini et al., 2022), and here it is expanded for resilience optimization. The model features transportation by ship, barge, train, truck and pipeline, and introduces the concept of expected carbon not stored (ECNS) under failures as a separate objective function. This objective function assesses the expected difference between the nominal supply (no failures) of CO_2 to the storage sites and the supply under a combination of failure events occurring during the transport of CO_2.

In fact, a resilient CO_2 transport network is key in guaranteeing the successful deployment of CCTS, particularly in Europe where CO_2 sources are spread across the continent, especially inland, while CO_2 storage sites are being developed mostly in the North Sea. The various CO_2 transport modes considered here (see Section 3 for a detailed technical description) have specific characteristics that make them most suitable in different operational regimes. Except for pipelines, the other transport modes are based on the use of containers that can be filled with CO_2 and loaded onto ship, barge, train or truck for transport. Although the CO_2 is kept in the containers from the source to the final delivery point (i.e., either a storage site or a pipeline entry point), the containers can be exchanged between infrastructure (e.g., from a barge to a ship terminal). It follows that these transport modes share the operational risks associated with the container itself, e.g., overpressure in the container leading to a CO_2 release in the atmosphere, which however do not generally result in a disruption of the transport connection itself. Such disruptions may be caused by traffic congestion for truck and rail transport, low-water levels or port congestion for barge transport, severe weather conditions mostly for ship transport. These may result in delays (from hours to days, or even weeks) and missed transport connections that could require the timely implementation and deployment of a backup connection. In the case of CO_2 transport via pipelines, day-, week-, or month-long disruptions could be caused by ruptures or material failures (e.g., due to corrosion), as those encountered with natural gas pipelines.

Risks associated to disruption of storage operations are not included for two reasons. First, there is still too little operational experience to build a statistically significant database of risks associated to storage that might lead to disruptions. Second, all current CCS operations connect one CO_2 source to one storage sink; there is no experience of the complexities associated to the operation of a storage hub that receives CO_2 from multiple sources, exploiting multiple supply chains. A resilience analysis of the latter situation, which is the relevant one in the long run, would be extremely interesting, but it is beyond the scope of this contribution.

Following our earlier analysis, we illustrate the optimal design of resilient CO_2 transport networks, or simply CO_2 networks, by investigating the decarbonization of the Swiss WtE sector via CCTS supply chains, which connect the Swiss WtE plants to potential CO_2 storage sites in the North Sea and in Switzerland.

The paper is structured as follows. Section introduces and defines the concept of resilient CCTS supply chains. Section 3 describes the system configuration used to perform the analysis. Section 4 formulates the multi-objective optimization model and describes the methodology and the metrics used to assess resilience. Section 5 investigates the trade-off between cost-optimal and resilient configurations. Finally, Section 6 summarizes findings and draws conclusions.

2. Resilient CCTS supply chains

The concept of resilience has been applied to a variety of fields for several decades (e.g., psychology, psychiatry, ecology, social science, economy, and engineering), thus resulting in different definitions (Carlson et al., 2012). The National Academy of Sciences defines *resilience* as the ability to absorb, adapt to, and/or rapidly recover from a potentially disruptive event, and *infrastructure resilience* as the ability to reduce the magnitude, impact, or duration of such a disruption (National Research Council, 2012). Therefore, the quantification of a system's resilience is strictly linked to its functionality.

The functionality of a CCTS supply chain infrastructure (hereafter simply denoted as CCTS supply chain), defined as the ensemble of assets and systems that constitute the network of CO_2 emitters, transport and storage, may be identified as its ability to permanently store the captured CO_2 during the time horizon of interest. Therefore, a resilient CCTS supply chain should be able to cope with disruptions and sustain its function of storing the captured CO_2, or to recover its functionality as quickly as possible if it is interrupted (Dell'Isola et al., 2020).

The importance of a resilient CCTS supply chain can be acknowledged from different perspectives, depending on the harm that may occur in case the infrastructure fails to operate properly. From a climate perspective, the interrupted functionality of the supply chain, which results in the release of CO_2 into the atmosphere, would prevent achieving net-zero-CO_2-emissions goals. From an economic perspective, a failure in delivering the CO_2 to the permanent storage site would result either in a tax being paid for the emissions caused or in a credit being missed in case the stored CO_2 would contribute to generate negative CO_2 emissions. Additionally, it is likely that in such a case the CO_2 emitter would incur in extra costs to be paid to, for example, the CO_2 transport or storage providers depending on legal and liability agreements in place. Furthermore, while statistics indicate that the risks of releasing CO_2 are lower than those of releasing natural gas or other hazardous substances (Duncan and Wang, 2014), releasing CO_2 should be avoided to prevent safety issues stemming from its high density as compared to air (Gale and Davison, 2004). Moreover, simply venting CO_2 to the atmosphere in case of failure might result in low public acceptance, whereas resilient CCTS supply chains would contribute to gain credibility and public confidence.

Based on the above considerations, we defined and investigate CCTS supply chain resilience by adopting two perspectives:

- From the *infrastructure resilience* standpoint, an infrastructure-resilient CCTS supply chain is able to store the CO_2 upon the occurrence of failures, while minimizing the total system costs;
- From the *environment resilience* standpoint, an environment-resilient CCTS supply chain is able to store the CO_2 upon the occurrence of failures, while minimizing the total system emissions.

Such resilient supply chains should be compared against cost-optimal supply chains, which release CO_2 into the atmosphere in case of failures.

The aforementioned perspectives should be taken into account when developing a wide CCTS supply chain infrastructure, and they highlight the need for resilient CCTS value chains able to operate even when failures occur.

Within the framework of CCTS supply chains, disruptions to the nominal operation could stem from a failure in capturing the CO_2 at the emissions sites (e.g. an industrial plant), from an interrupted connection between the capture site and the storage site, from delays in transporting the CO_2, or from an interruption in the operation of the storage facilities. When looking at CO_2 transport, and especially at CO_2 pipelines, the relevant failure mechanisms are similar to those of natural gas pipelines. More specifically, both CO_2 and natural gas pipelines are fabricated from carbon steel, both are installed using similar equipment and practices (Duncan and Wang, 2014), and both are subject to potential internal corrosion damage, as well as potential failures due to excavation issues (Sim et al., 2014).

Whereas we do not perform a risk assessment, and we base our study on previous analyses that quantify the failure rates of the different components of a CCTS supply chain (Zep, 2011; Jo and Crowl, 2008; Duncan and Wang, 2014; Expert interviews, 2021), we propose strategies to increase the resilience of CO_2 networks, hence focusing on the transport step of CCTS supply chains. Such strategies may include, for example, installing backup transport connections along the CO_2 network, and may favor one transport technology with respect to another when comparing resilient and cost-optimal supply chains. We focus on one particular strategy, namely using backup transport connections. Though this is only one among various options for designing resilient networks, it sets the ground for a comprehensive resilience analysis of CCTS supply chains by defining and quantifying resilience indicators and by investigating trade-offs between resilience, costs and CO_2 emissions.

3. System description

The system considered in this work is illustrated in Fig. 1 and is described in details in Section 2 of Becattini et al. (2022). The CO_2 is captured at the Swiss WtE plants and is transported via a multi-modal network to the storage sites, where it is permanently sequestered underground. The resulting CCTS supply chains are represented as a network, where the nodes consist of the CO_2 capture sites, the CO_2 storage sites and the transport exchange sites. Below, we provide a short description of the nodes and of the available connections, focusing on CO_2 transport.

CO_2 capture sites. The thirty Swiss WtE plants emit a total of ca. 4.5 $MtCO_2$/year, about half of which being generated from biogenic sources of carbon (Verband der Betreiber Schweizerischer Abfallverwertungsanlagen (VBSA), 2019, 2021).

CO_2 storage sites. Two CO_2 storage sites are considered, namely the Northern Lights CO_2 storage hub currently being developed in the North Sea (Equinor, 2019), and a hypothetical storage site in Switzerland.

The Northern Lights storage site is assumed to be available during the entire time horizon. In contrast, the Swiss CO_2 storage site is assumed to become operational in 2040, after 15 years from the first installation of the network. A sensitivity analysis of the cost-optimal design on the starting date of operation of the Swiss storage site is illustrated in Figure 10 of Becattini et al. (2022).

Transport exchange sites. Two additional sites in Rotterdam and in Basel, with neither capture nor storage capabilities, are included to function as transit terminals. The Rotterdam harbor connects continental Europe to the Northern Lights CO_2 storage hub. The transport exchange site located in Basel is combined with the WtE plant into a single node; it serves as collection point for the Swiss CO_2 and is directly linked to the Rotterdam exchange site.

Transport exchange sites are setup to handle the transportation switch between truck, rail, barge and pipeline. Compressors are assumed to be available at each node (i.e., capture, storage, and transport exchange sites) to provide any conditioning work required.

Multi-modal transport. All nodes can be connected via multiple modes of transport:

- Truck connections are available between sites located in Switzerland (WtE plants and Swiss storage site) and between Basel, Rotterdam and the onshore terminal of the Northern Lights storage site. The CO_2 is transported in liquid form at a pressure of 22 bar and at a temperature of −35 °C using an ISO-tank container with unit capacity of 20 tCO_2, see e.g., COMTECSWISS (2021). The length of each truck connection corresponds to the shortest available road connection.
- A rail connection is included between Basel and Rotterdam. In addition, certain WtE plants are connected to Basel via railway lines departing from their own private stations located on site, which allow them to load CO_2 directly onto the container attached to a train. As for truck connections, the CO_2 is transported in liquid form using ISO-tanks (at the same pressure and temperature as for truck-based transport), which can be loaded directly on rail cars. The length of each rail connection corresponds to the shortest available one.
- Greenfield pipelines can be installed between the WtE plants, the exchange transport sites, and the storage sites. The CO_2 is transported in dense phase at 110 bar and ambient temperature. To account for the complexity and the duration of pipelines construction, we allow the size (i.e., diameter) of pipelines to be expanded only after at least ten years from their installation for a given configuration. Realistically, this translates into the constraint that, if installed, a pipeline cannot be dismantled or replaced with a smaller one during the considered time horizon. For the sake of simplicity, the length of each pipeline connection corresponds to the shortest distances between two sites. Although CO_2 pipelines are not new, this transport type requires attention to be paid to design factors, overpressure protection, and leak detection, especially when occurring through populated areas (Metz et al., 2005).
- Barge and ship connections are available between Basel and Rotterdam (along the Rhine river), and between Rotterdam and the offshore terminal of the Northern Lights storage site (located nearby the city of Bergen), respectively. Although ships specifically designed to transport CO_2 in liquid form for CCTS applications are at an early stage of development (Aspelund et al., 2006; Kler et al., 2016; Al Baroudi et al., 2021), we assume that ISO-tank containers are loaded onto conventional cargo ships (Bjerketvedt et al., 2020). As for truck- and rail-based transport, CO_2 is transported in liquid form at a pressure of 22 bar and at a temperature of −35 °C. Barge and ship share similar costs and emission values and mainly differ in terms of their average travel speed.

CO_2 emissions pathway. The optimal planning, development and roll-out of CCTS supply chains must comply with the linear emissions reduction (LER) pathway introduced earlier (Becattini et al., 2022). Here, the total CO_2 emissions of the chain decrease linearly from the current value in 2025 (when the CO_2 network starts being installed) to the minimum value attainable in 2050 (after 25 years from the first installation of the network).

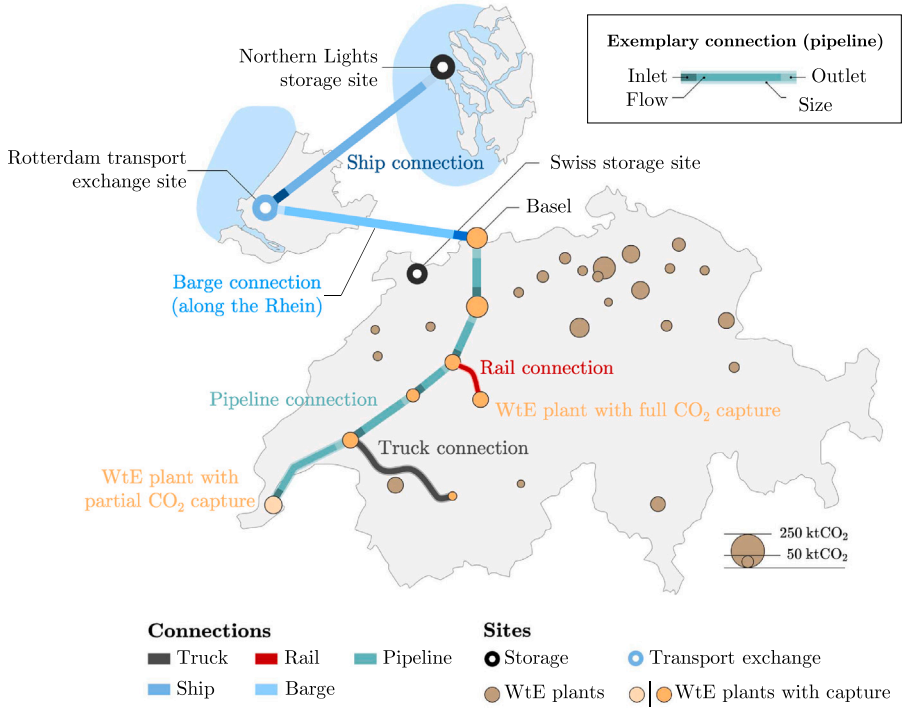


Fig. 1. Scheme of the system configuration showing: the WtE plants in Switzerland, the CO_2 storage sites (i.e., the Northern Lights site and the hypothetical site in Switzerland), the transport exchange sites (i.e., Rotterdam and Basel; note that for the latter the transport exchange site is combined with the WtE plant into a single node), and a few illustrative connections (not referring to any optimal design). For illustrative purposes, some WtE plants are represented as equipped with a CO_2 capture unit (yellow dots, with the light-shaded yellow dot indicating a CO_2 capture unit installed on a fraction of the exhaust gases, i.e. the recovery rate is smaller than 90%). The inset on the top right exemplifies the representation of a connection with the light-shaded area indicating the size of the installed connection (i.e., maximum capacity installed), in contrast to the darker area indicating the actual CO_2 flow that is transported (i.e., deployed capacity); the dark-shaded and the light-shaded ends indicate the inlet and the outlet of the connection, respectively. The line thickness is proportional to the CO_2 flow/size of the connection.

3.1. *Backup connections and resilient* CO_2 *networks*

The resilience of CO_2 capture and storage are not considered in this study. We refer to resilient CCTS supply chains that are resilient in terms of CO_2 transport, as illustrated in Fig. 2.

A CO_2 transport network can be designed to be cost-optimal only (Fig. 2a) or to be cost-optimal and resilient (Fig. 2b). In this work, we refer to the latter case simply as *resilient*; a resilient CO_2 network is designed and operated in a way that minimizes costs while complying with a specified level of resilience (see Section 4.2).

We define the *nominal state* as the state of a supply chain where no failure occurs (Fig. 2c and d). A cost-optimal supply chain is designed to be cost-optimal with respect to the nominal state only. In contrast, a resilient supply chain is designed to be cost-optimal when accounting for both the nominal state and all possible failure states; it features *backup connections* that allow it to withstand failures, but might be unused in the nominal state (Fig. 2f).

When a failure occurs in a cost-optimal supply chain (Fig. 2e), e.g. a pipeline stops being available due to a corrosion damage, the amount of CO_2 corresponding to the failed connection is lost to the environment. In contrast, when a failure occurs in a resilient supply chain (Fig. 2f), CO_2 is re-routed towards the storage site through alternative backup connections, i.e. trucks that replace the pipeline, and is eventually sequestered underground.

All transport modes mentioned above, namely truck, rail, pipeline, barge and ship, can be deployed as backup connections, with each featuring specific logistic considerations.

- A backup pipeline would be constructed between nodes to create a meshed network or a redundant connection. This would sit idle until a failure occurs, when CO_2 is re-routed through the backup connection, cutting out the nominal pipeline. It is worth noting that cost-optimal pipelines could potentially be oversized to accommodate additional volume that would be transported once the failure is resolved, to deliver the amount of CO_2 not transported during failures. However, this implies being able to temporary store CO_2 in buffer tanks during failures. While this is possible in principle, it is not investigated here; the design of temporary storage, as well as detailed considerations concerning the minimum and maximum flow rate that can be feasibly transported through pipelines require a finer time resolution than the one considered here.
- In contrast to pipelines, backup trucks can be rented and used in different places at different points in time. However, they would be parked in a warehouse and unused until a failure occurs. In this work we do not consider neither parking costs, nor potential revenues for non-related uses during the idle time, but we simply consider the cost of reserving the trucks to have them available when the failure occurs.
- Similar considerations to trucks apply to the other transport modes carrying ISO-tank containers, namely rail, barge and ship.

Finally, in investigating system resilience, we refer to the distinct concepts of *risk* and *recovery* after failure. Assuming the failure of a connection, the concept of risk refers to the state of the system immediately after failure (red time instant in Fig. 2f), when the connection is failed and the CO_2 is not yet re-routed to a backup connection. The concept of recovery (orange time interval in Fig. 2f) refers to the state of the system during the entire time interval of interest; this includes the recovery time, which is specific for each transport mode (see Table A.1) and during which the backup connection transports the CO_2 re-routed from the failed connection, as well as the post-recovery period, where the system can operate in normal conditions and the backup connection is not required anymore. For all transport modes, no lag time is considered for activating the backup system.

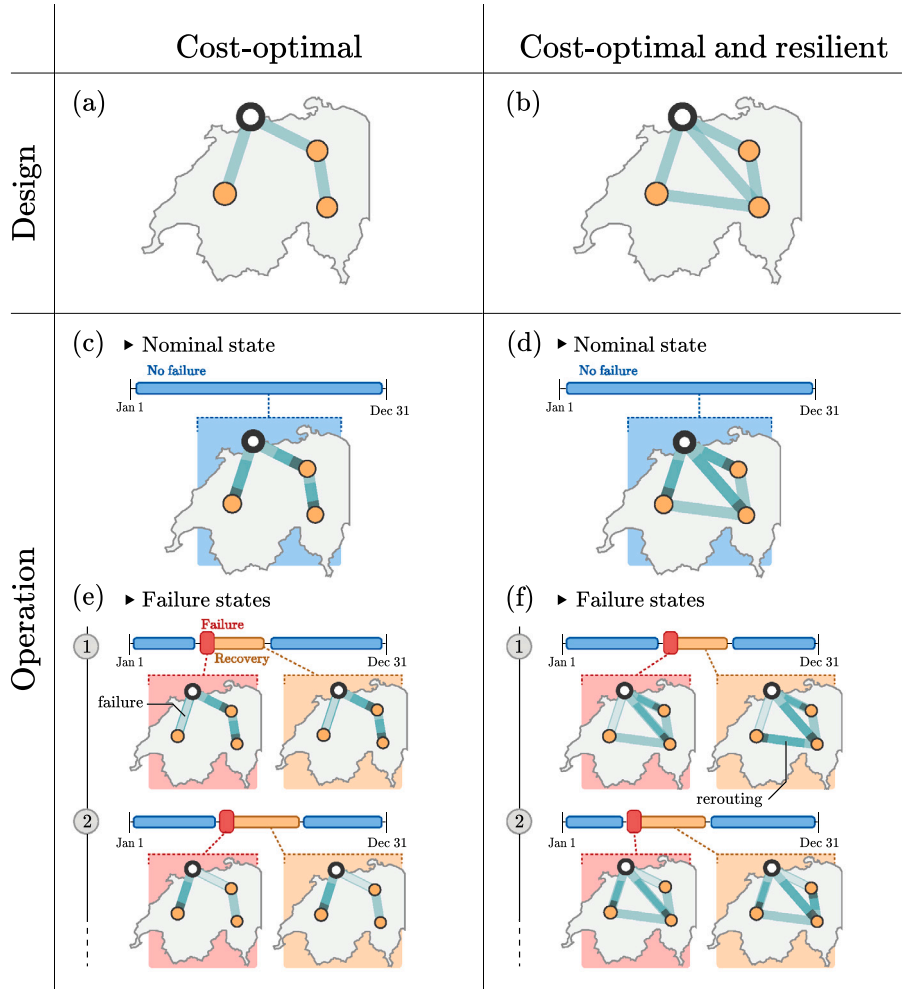


Fig. 2. Illustration of the definitions used to determine and compare the minimum-cost and the maximum-resilience designs of CCTS supply chains.

4. Formulation of the optimization problem

The optimal design of CCTS supply chains is determined by solving an optimization problem, which minimizes the total costs and maximizes the resilience of the system, while complying with the CO_2 emissions scenario described in Section 3.

The cost-optimal supply chain is presented elsewhere (Becattini et al., 2022), and it is an input data of the optimization problem. This consists of: (i) location and size of the installed CO_2 capture technologies, and (ii) type, size and flow of the installed transport technologies. Starting from this, additional (or backup) connections are determined that allow the supply chain to transport CO_2 in the event of a cost-optimal connection not being available (to make the cost-optimal solution resilient against failures). Therefore, the optimization algorithm (i) determines the most cost-effective strategy, in terms of backup connections, to make the cost-optimal solution resilient, and (ii) quantifies the cost of such a strategy. We consider the failures of all connections present in the cost-optimal solution, with one single failure occurring at a time. The probability of more than one component failing simultaneously is assumed to be zero (rare event approximation). In other words, the optimization strategy determines the *(N-1)-resilient* design for CO_2 networks, i.e. a design that is cost-optimal in nominal conditions, and that is always able to deliver CO_2 to the storage site should one connection at a time become unavailable.

The optimization problem is formulated as a MILP and builds upon the framework developed previously (Gabrielli et al., 2018a,b, 2019, 2020a; Becattini et al., 2022). Alongside the cost-optimal design, input data to the optimization problem are: (i) location and current CO_2 emissions of WtE plants, (ii) location and capacity of storage sites, (iii) performance and costs of capture, storage and transport technologies, (iv) availability and reliability of transport technologies, and (v) time-dependent CO_2 emissions pathway that the overall WtE sector needs to follow. Constraints include technology performance, energy balance and flow conservation.

We define the set of all possible states as S, which includes the nominal and the failure states. The number of possible states is defined by the number of components of the CCTS supply chain subject to failure. Here, these are the transport connections only.

4.1. Economic indicators

The economic performance of the CCTS supply chain is assessed via the levelized cost of stored carbon (LCSC) (Becattini et al., 2022). This relies on the present value of the total cost of the CCTS infrastructure in year τ, J_τ, which is expressed as:

$$J_{s,\tau} = \sum_{t=0}^{\tau} \frac{C_{s,t} + O_{s,t} + M_{s,t}}{(1+r)^t} \quad (1)$$

where s is a state within the set of all possible states of the supply chain, S; $C_{s,t}$, $O_{s,t}$ and $M_{s,t}$ are the annual capital, operation and maintenance cost of the CCTS supply chain in state s; r is the discount rate; $\tau \leq T$, with T being the total number of years considered for the analysis. Note that the capital cost is annualized via the discount rate r, so that the capital, operation and maintenance costs all refer to the common basis

of one year. Furthermore, Eq. (1) indicates that all costs are discounted based on the point in the future in which they occur.

The levelized cost of stored carbon in state s and year τ is obtained by dividing the present cost, $J_{s,\tau}$, by the present value of the amount of CO_2 stored, $U_{s,\tau}$ (Rubin et al., 2013):

$$\mathrm{LCSC}_{s,\tau} = \frac{J_{s,\tau}}{\sum_{t=0}^{\tau} \frac{U_{s,t}}{(1+r)^t}} \tag{2}$$

where $U_{s,t}$ is the amount of CO_2 physically stored during year t in state s. Eqs. (1) and (2) are written for all states $s \in S$.

4.2. Resilience indicators

The resilience of a CCTS supply chain is defined as its ability to mitigate the effect of failures on its function to successfully capture, transport and permanently store CO_2 during a time horizon of interest. It is based on the concept of expected carbon not stored (ECNS), which quantifies the expected difference in the amount of CO_2 stored when the state of the system has changed (e.g., due to failures) compared to nominal conditions. The ECNS is quantified by looking at the entire time horizon of interest and considers the possibility to recover from failures after a given repair time, δ_l, specific for technology l. The ECNS in year τ is expressed as:

$$\mathrm{ECNS}_{\tau} = \sum_{s \in S} \pi_s \mathrm{CNS}_{s,\tau} \tag{3}$$

where,

$$\mathrm{CNS}_{s,\tau} = \sum_{t=0}^{\tau} \frac{U_{\mathrm{nom},t} - U_{s,t}}{(1+r)^t} \tag{4}$$

where π_s is the probability of the system to be in state s (with $\sum_{s \in S} \pi_s = 1$); $U_{\mathrm{nom},t}$ and $U_{s,t}$ are the amounts of CO_2 stored during year t in the nominal state and in state s, respectively.

All failure states are characterized by one failed cost-optimal connection. In this case, π_s is computed via an event tree where only one component is failed at a time (Billinton and Allan, 1992; Hadjidimitriou et al., 2021). Denoting with $\mathcal{K}$ the set of cost-optimal connections, the probability of the system to be in the failure state is expressed as:

$$\pi_s = \sigma_i \prod_{j \in \mathcal{K}, i \neq j} (1 - \sigma_j) \tag{5}$$

where i is a connection being failed and j is a connection being operational; σ_i is the probability of connection i being unavailable during the time horizon of interest, and $1 - \sigma_j$ is the probability of connection j being operational during the time horizon of interest. The probability of the system to be in nominal conditions is the complement to 1 of the sum of the single-failure probabilities; states where multiple failures occur at the same time are neglected (rare event approximation). We assume the failure probabilities of the components to be independent of each other and of time.

The set of connections, $\mathcal{K}$, consists of different transport technologies, $l \in \mathcal{L}$, which connect nodes i and $i' \in \mathcal{N}$. With this notation, we define the failure probability σ_i for all technologies and for all connections between nodes:

$$\sigma_{i,i',l} = \delta_l \lambda_l L_{i,i',l} \Delta t \tag{6}$$

where λ_l is the failure rate per unit time and distance (per year, per km) of technology l; $L_{i,i',l}$ is the length (in km) of the connection between nodes i and i' when connected through technology l; δ_l is the yearly time fraction needed to repair technology l ($\delta_l = \delta_l/365$, with δ_l expressed in days); Δt is the time horizon of interest (one year). The values of all input data used for the optimization are reported in Table A.1 in Appendix A (resilience) and in the Appendix of Becattini et al. (2022).

Based on the ECNS, the resilience of CCTS supply chains is assessed by taking the cost-optimal (c-opt) supply chain as a reference. More specifically, the resilience of the system, $R \in [0, 1]$, is expressed as:

$$R_{\tau} = 1 - \min\left(1, \frac{\mathrm{ECNS}_{\tau}}{\mathrm{ECNS}_{\text{c-opt},\tau}}\right) \tag{7}$$

where ECNS_{τ} is the ECNS for any resilient supply chain computed through Eq. (3), and $\mathrm{ECNS}_{\text{c-opt},\tau}$ is the ECNS of the cost-optimal supply chain in year τ. R takes the value 0 when ECNS_{τ} is greater than or equal to $\mathrm{ECNS}_{\text{c-opt},\tau}$ (i.e. the supply chain does not improve the resilience of the cost-optimal solution); it takes the value 1 when the CCTS supply chain is resilient to all events involving at most K simultaneous failures.

The cost of resilience is determined by computing the expected value of LCSC (ELCSC), when considering failures:

$$\mathrm{ELCSC}_{\tau} = \sum_{s \in S} \pi_s \mathrm{LCSC}_{s,\tau} \tag{8}$$

Similarly, the expected CO_2 emissions of the supply chain are determined by considering the CO_2 emissions of all possible system states and their corresponding probabilities.

4.3. Model constraints

The constraints of the optimization problems include all those presented in the Supporting Information (Section 1) of Becattini et al. (2022), and are written here for all states $s \in S$. In addition to these, the CO_2 flow within a failed connection is constrained by the value of the nominal flow and the time required for repair. The CO_2 flow transported by technology l, between nodes i and i', during year t, in system state s, $G_{s,i,i',l,t}$, is constrained by:

$$G_{s,i,i',l,t} \leq (1 - \delta_l) G_{\mathrm{nom},i,i',l,t} \tag{9}$$

All states are considered simultaneously through a single-stage optimization. Here, we do not consider any lag time, or re-routing time, for the backup connections due to lack of data. One way of considering the re-routing time would be including it in the parameter δ. However, a detailed description of this and other real-world market mechanisms (e.g., impacting the availability and resilience of backup connections) would likely require a time resolution finer than one year, and goes beyond the scope of this study.

4.4. Addressing computational complexity

Accounting for network resilience, and specifically implementing the calculation of ENCS, leads to an exponential build up of decision variables, hence of the complexity of the optimization problem. As an example, for a transportation network of n nodes with a fully connected network, the total number of components increases on the order of $\mathcal{O}(n(n-1)) = \mathcal{O}(n^2)$ (Billinton and Allan, 1992). If we consider every individual component failure over a single year, the number of decision variables used for resilience is on the order of $\mathcal{O}((n^2-1)n^2) = \mathcal{O}(n^4)$. For a system of 30 nodes, this leads to around 800,000 new variables for a single transportation network.

To handle such complexity, the following two approaches are leveraged.

- Pruning the connectivity matrix. Connections between distant nodes can be removed using a variety of techniques. An example is provided by Morbee et al. where Delaunay triangulation is used to pre-select available pipeline connections (Morbee et al., 2012).
- Reducing the number of nodes within a single optimization procedure. This can be done with clustering techniques, such as k-medoids or agglomerative hierarchical clustering (Emmons et al., 2016; Fortenbacher et al., 2018). Here the latter is chosen, where the network is subdivided into multiple sub-networks and solved separately, similar to energy transmission and distribution networks (del Granado et al., 2018). The optimal sub-networks are then combined to approximate the global optimal network.

All optimizations are formulated in Matlab (MathWorks, 2019), using the modeling and optimization toolbox Yalmip (Löfberg, 2004). They are solved by using the commercial software CPLEX Optimization Studio by IBM (IBM, 2019), set up to have a relative MIP gap of 1%. All optimizations are run on a computer featuring a 12-core processor Intel(R) Xeon E5-2680v3, with a peak clock speed of 3.3 GHz and 64 GB of installed RAM.

5. Optimal trade-off between cost and resilience

The cost-optimal design of CCTS supply chains in nominal conditions (neglecting failures) has been investigated earlier for different CO_2 emissions reduction pathways (Becattini et al., 2022). Here, we study the (N-1)-resilient design of CCTS supply chains, as well as the trade-off between cost-optimal and resilient solutions, for the LER pathway defined in Section 3.

Maximizing the resilience of the CCTS supply chains fosters the installation of backup connections, which support the operations of the network during failures. The two perspectives on CCTS supply chain resilience introduced in Section 2 entail two design solutions, namely:

- Infrastructure resilience. The total expected cost of the supply chain is minimized when considering failures, while ensuring a specified value of resilience (i.e., of ECNS). The expected emissions of the supply chain when considering failures are accounted for. This is relevant from the perspective of infrastructure planners and operators, as it provides an estimate of the minimum costs incurred to attain a resilient CCTS supply chain.
- Environment resilience. The total expected CO_2 emissions of the supply chain are minimized when considering failures, while ensuring a specified value of resilience (i.e., of ECNS). The expected costs of the supply chain when considering failures are accounted for. This is relevant from the perspective of the environment, as backup connections might eliminate the risk of not storing carbon (possibly connected to carbon credits or certificates) at the expenses of additional emissions.

5.1. Cost-ECNS Pareto front

The trade-off between cost and resilience is exemplified with reference to a reduced CO_2 network. This is obtained by clustering all the WtE plants into three hubs via agglomerative hierarchical clustering (with the exception of the Basel plant that belongs to the Basel hub) (Rokach and Maimon, 2005; Pedregosa et al., 2011). The optimization problem is solved for one year, i.e. $\tau = 0$ in Eqs. (1)–(4), (7) and (8), and it is constrained so that all WtE plants capture CO_2, which is then transported and stored. The emissions of the individual plants are summed and assigned to their corresponding hubs (see Table B.4 in Appendix B). The resulting network consists of (i) the North Sea storage site, (ii) Rotterdam, (iii) Basel (257ktCO_2/y), (iv) Geneva (230ktCO_2/y), (v) Giubiasco (190ktCO_2/y), and (vi) Luzern (3.8MtCO_2/y). The Basel hub can be connected directly to the North Sea storage site. The high concentration of emitters in the North-East of Switzerland explains the imbalance in emissions amongst the hubs. This imbalance serves the purpose of creating connections with significantly varying flow rates. The cost-resilience Pareto front is constructed by minimizing the supply chain costs for fixed values of ECNS, via an ϵ-constraint method (Mavrotas, 2009). The minimum and maximum values of costs are determined via two single-objective optimizations for cost and resilience, respectively. We examine the Pareto fronts for infrastructure resilience, i.e. the total cost of the supply chain is minimized while ensuring a fixed value of resilience, for a different availability of transport technologies. These are shown in Fig. 3, where resilience is quantified through R_0 and the cost through $LCSC_0$ (single-year analysis, $\tau = 0$). For reference, the cost-optimal solution results in ECNS = 0.22 MtCO_2 per year ($R_0 = 0$ - about 5% of the captured CO_2 does not reach the storage site), whereas $R_0 = 1$ corresponds to zero ECNS under single failure.

The shape of the Pareto fronts confirms that cost and resilience are competitive objective functions, and that the resilience of the cost-optimal network can only be increased at some additional cost. The network achieves maximum resilience ($R_0 = 1$) at an LCSC increase of: 43% for the pipeline-only network (○); 35% for the pipeline and rail network (◇); 27% for the pipeline, ship and barge network (▽); 6% for the pipeline and truck network (□), which overlaps (and covers in the figure) the network with all modes of transport (⋆). In other words, the combination of pipelines and trucks is sufficient to obtain the best cost and resilience performance. Trucks benefit from low investment costs, high speed of transport and wide availability, and therefore outperform all other modes of transport when adopted as a backup solution to increase the infrastructure resilience of a pipeline-based cost-optimal design. It is worth noting that:

- Trucks are not selected in the cost-optimal design due to their high operation costs (Becattini et al., 2022); however, operation costs are negligible when deploying trucks as backup connections, as they need to run only during the time where pipelines are unavailable.
- Following the rare-event approximation, we consider one failure at a time; therefore, the same trucks can act as multiple backup connections, in different geographical locations and at different times, and the amount of trucks required is determined by the largest connection. This is different when considering pipeline backup connections, as additional pipelines should be installed along the cost-optimal connections. For the considered system, the largest connection is the one between Basel and the North Sea storage site (see Fig. 4f). Considering a distance of about 2200 km, a pipeline recovery time of 20 days, an average travel speed of 35 km/h, and a truck (i.e., isotainer) capacity of 20 tCO_2, the backup connection would consist of about 3000 trucks, transporting about 220,000 tCO_2 during 20 days.
- While we account for the capital cost of the trucks, for the sake of simplicity we do not consider additional costs that might incur when keeping a large fleet of trucks parked and non-operational for long periods of time.

The CCTS configurations of some selected points on the Pareto fronts are shown in Fig. 4, which provides the following insights in terms of resilience measures:

(a) Cost-optimal ($R_0 = 0$). The cost-optimal network is entirely based on pipelines, and is obtained also when all transport modes are available. The cost-optimal network routes through the shortest path to the North Sea storage, adopts no backup connections, and releases CO_2 to the atmosphere in case of failure.

(b) Local resilience with pipeline only ($R_0 = 0.10$ at a 5% cost increase). The pipeline connection from Giubiasco to Luzern is placed side-by-side with one from Giubiasco to Basel with increased capacity. An additional backup pipeline connection is installed from Luzern to Giubiasco to redirect the CO_2 captured in Luzern in case of failure along the Luzern-to-Basel connection.

(c) Global resilience with pipeline only ($R_0 = 0.90$ at a 32% cost increase). The local topology is reverted back to the cost-optimal topology and a secondary backup path through Rotterdam to the North Sea storage site is installed to support the connection from Basel to the North Sea.

(d) Local and global resilience when adding rail and ship transport ($R_0 = 1$ at a 35% cost increase). Pipelines are installed to be resilient against local failures. The connection from Basel to the North Sea storage site is redirected through Rotterdam, and two additional backup connections are installed, namely a rail connection from Basel to Rotterdam and a ship connection from Rotterdam to the North Sea storage site, to be resilient against global failures.

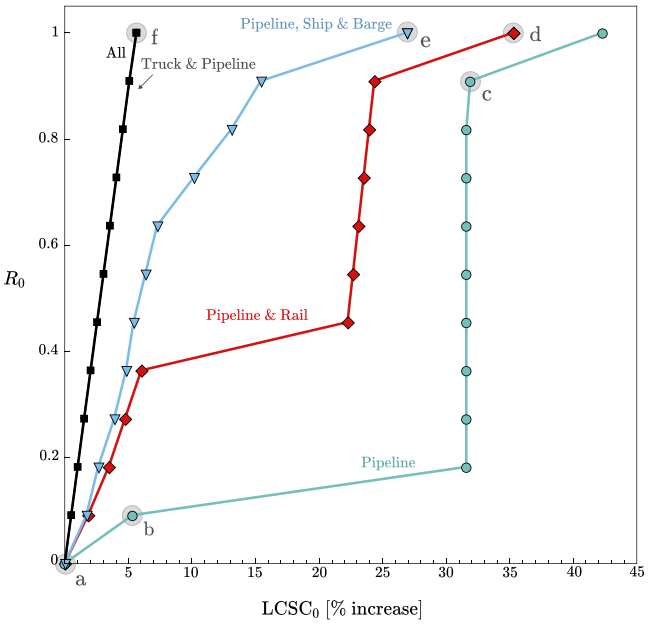


Fig. 3. Infrastructure-resilient CCTS supply chains for the reduced network used to illustrate the cost-resilience tradeoffs in a single-year resilience analysis ($\tau = 0$). Pareto fronts of $LCSC_0$ and R_0 for different available transport technologies: pipeline only (○); pipeline and rail (◇); pipeline, ship and barge (▽); pipeline and truck (□); all available technologies, namely pipeline, truck, rail, ship and barge (⋆ - coinciding and underneath □ in the figure). For reference, the cost-optimal solution results in ECNS = 0.22 $MtCO_2$ per year ($R_0 = 0$).

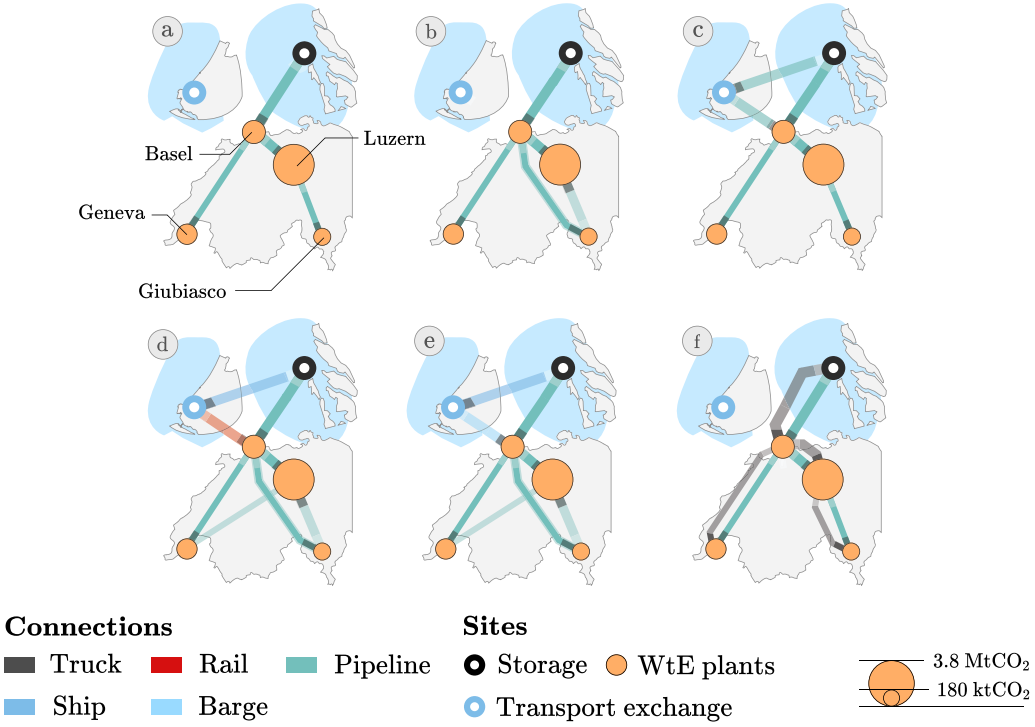


Fig. 4. Infrastructure-resilient CCTS supply chains for a simple network used to illustrate the cost-resilience tradeoffs for a single-year resilience analysis ($\tau = 0$). Schematic representation of supply chain structures for different available transport technologies and for different values of resilience. Connections represented with a shaded color indicate backup solutions.

(e) Local and global resilience when adding barge transport ($R_0 = 1$ at a 27% cost increase). The rail connection between Basel and Rotterdam is replaced by a barge connection, which allows to reduce the cost of resilience.

(f) Local and global resilience when adding truck transport ($R_0 = 1$ at a 6% cost increase). A backup truck network is installed to support during local and global failures of the pipeline network. This is the (N-1)-infrastructure-resilient optimal solution, as it maximizes the resilience of the cost-optimal network against single failures in the most cost-effective fashion.

Overall, we can conclude that:

- The inclusion of multiple modes of transportation confers an overall advantage to the CCTS supply chain. Different modes

perform best within certain ranges of volume of CO_2 transported and of distance. Combined, their properties can be leveraged to design a resilient network at different scales. In this case, the low investment costs, high speed of transport and wide availability of trucks make them the best backup technology, allowing for a drastic reduction in the cost of resilience compared to the other modes of transport (see Figs. 3 and 4f). This conclusion is, however, subject to the values of the input parameters used in the optimization problem. It might change when considering additional types of costs (for example, the costs of keeping a large fleet of trucks, 3000 in this study, parked and non-operational for long periods of time). The sensitivity of the chain design to the values of the input parameters is investigated in Section 5.3.

- Installing backup connections to support the longest- and highest-volume connections offers the best resilience improvement, because the probability of failure increases with the distance traveled and because the importance of an edge increases with its flow. This occurs in radial networks, where the flow often ends up being concentrated in a few edges, or in a single edge, before reaching the storage site.

5.2. Resilient CCTS supply chains

Let us now study the resilience of the full CCTS supply chain, i.e. for the CO_2 network connecting the entire Swiss WtE sector described in Section 3. We perform a single-year resilience analysis (i.e. we quantify R_0) to determine the optimal CCTS supply chains in terms of (N-1)-resilience for the years 2030, 2040 and 2050. The single-year optimizations are constrained in such a way that the cost-optimal solutions match those obtained through the minimum-cost, multi-year optimization in Becattini et al. (2022).

Due to the large number of WtE plants present in the cost-optimal CCTS supply chain in 2030, 2040 and 2050, two different methods are adopted to handle the computational complexity resulting from the increased number of possible connections and failures (see Section 4.4). For all years, Delaunay triangulation is used to prune the connectivity matrix and to keep connections between close neighbors only. For the 2050 supply chain, where the entire network of WtE plants is part of the CCTS supply chain, the first method alone is not sufficient to handle the complexity of the optimization problem. Applying Delaunay triangulation still results in 468,000 decision variables (93,600 binary and 374,400 continuous variables); when solved on a computer featuring a 12-core processor Intel(R) Xeon E5-2680v3, with a peak clock speed of 3.3 GHz and 64 GB of installed RAM, the problem reaches a MIP gap of about 50% after three days (hence not converging to the required threshold of 1%). Therefore, clustering is utilized to further reduce the dimensionality of the problem. The network is subdivided into smaller networks, which undergo their own resilience optimization. The resilient sub-networks are then reassembled. This procedure prevents nodes belonging to different sub-networks from forming backup connections.

Fig. 5 shows the expected LCSC (left-hand side bars and y-axis) and the expected CO_2 emissions (right-hand side bars and y-axis) for the cost-optimal (gray), the infrastructure-resilient (blue), and the environment-resilient (green) designs. The three cases are reported for years 2030, 2040 and 2050. The reduced network presented in Section 5.1 is also reported for reference. The configurations of selected CCTS supply chains are presented in Fig. 6.

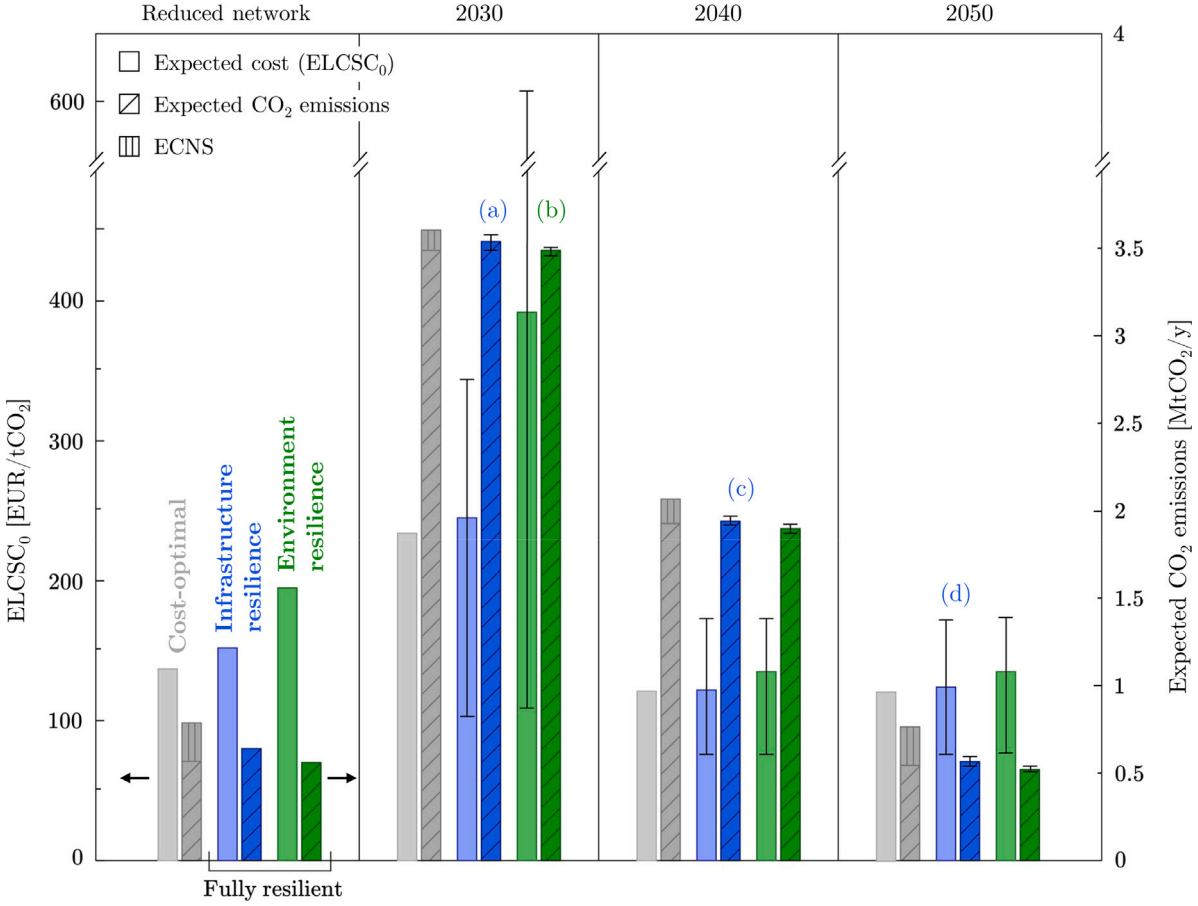


Fig. 5. Resilient CCTS supply chains for the Swiss WtE sector complying with a LER pathway in 2030, 2040 and 2050. Expected LCSC (left-hand side bars and y-axis) and expected CO_2 emissions (right-hand side bars and y-axis) for the cost-optimal (gray), the infrastructure-resilient (blue), and the environment-resilient (green) designs. Sensitivity ranges are indicated by the black whiskers and are obtained as described in Section 5.3. (a), (b), (c), and (d) refer to the supply chain configurations shown in Fig. 6.

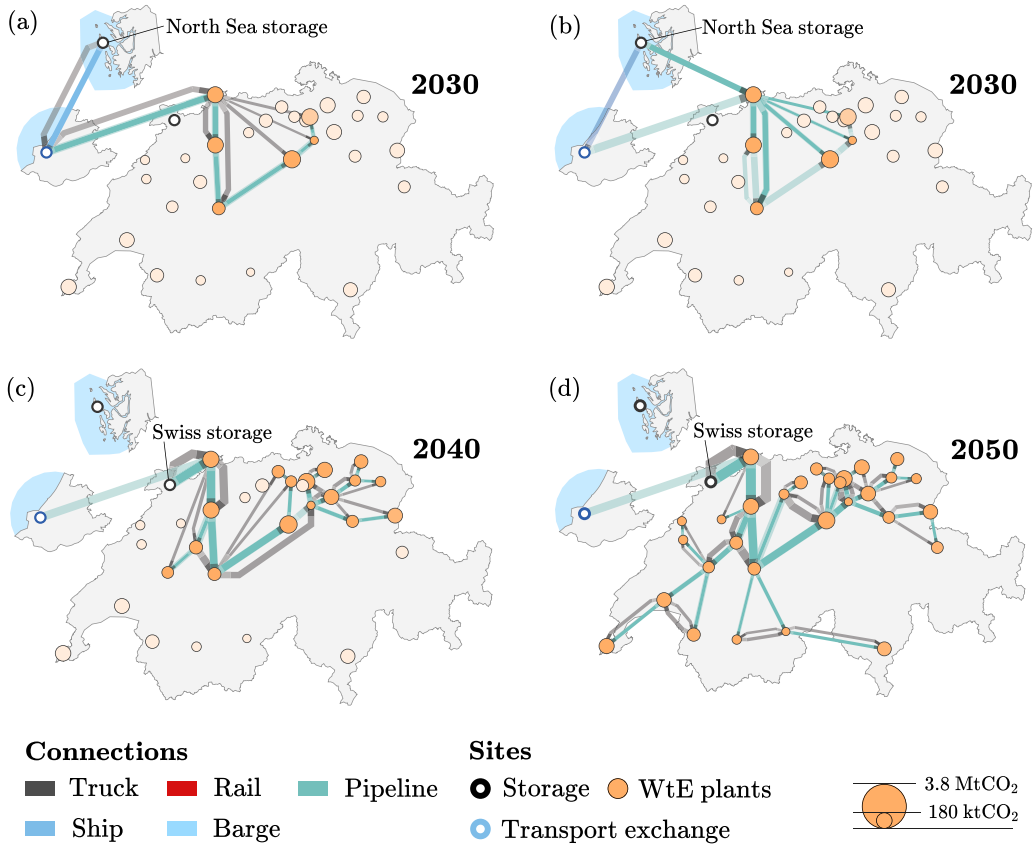


Fig. 6. Resilient CCTS supply chains for the Swiss WtE sector complying with the LER pathway in 2030, 2040 and 2050. Schematic representation of the (N-1)-resilient design, based on the cost-optimal, multi-year solution. Infrastructure-resilient CCTS supply chains shown for years 2030 (a), 2040 (c) and 2050 (d), and environment-resilient CCTS supply chain shown for 2030 (b). The Swiss storage site is available in 2040 (b) and 2050 (d). Shaded colors indicate connections that are either no longer utilized (e.g. the pipeline connection between Basel and Rotterdam in 2040 and 2050), or that are backup solutions (e.g. the truck connections in 2030, 2040 and 2050).

In 2030, the CCTS supply chain maximizing infrastructure resilience (blue bars in Fig. 5) is obtained at a nominal cost of 243 EUR/tCO_2 (+4% with respect to the cost-optimal solution), due to the additional investment costs required for the backup truck connections; when considering the expected costs in failure conditions, the total cost of the CCTS supply chain becomes 245 EUR/tCO_2 (+5%). This allows to avoid venting CO_2 into the atmosphere in the case of failures, hence reducing the expected CO_2 emissions of the cost-optimal supply chain by 50 ktCO_2 per year (about 1.5% of the CO_2 emissions of the WtE sector in 2030).

The CCTS supply chain maximizing environment resilience (green bars in Fig. 5) comes at a much higher cost of 390 EUR/tCO_2 (+67% with respect to the cost-optimal solution), due to the additional investment costs required for the backup pipeline connections, which must be installed in addition to the cost-optimal connections (see Fig. 6b).

Infrastructure resilience is obtained via a truck network (see Fig. 6a), which allows to minimize the costs, whereas environment resilience is obtained via redundant pipelines (see Fig. 6b), which allow minimizing the CO_2 emissions. The latter allows reducing the expected carbon emissions with respect to the cost-optimal solution by 100 ktCO_2 per year (about 3% of the CO_2 emissions of the WtE sector in 2030), but results in a drastic increase of costs.

The cost discrepancy between infrastructure and environment resilience becomes less pronounced in 2040 and 2050; this is because the differences between truck and pipeline are leveled off thanks to the assumed availability of the Swiss storage site, hence to the shorter distance for CO_2 transport, and to a greater utilization of the installed pipelines. The impact of the resilient designs on the CO_2 emissions of the supply chains increases from 2030 to 2050, as it depends on the amount of CO_2 transported. The infrastructure-resilience supply chain allows reducing the expected carbon emissions with respect to the cost-optimal solution by 200 ktCO_2 per year (about 25% of the CO_2 emissions of the WtE sector in 2050). Additional 50 ktCO_2 (up to 32% of the CO_2 emissions of the WtE sector in 2050) can be saved through the environment-resilience supply chain.

For all cases, the expected operation costs and emissions do not differ significantly from the nominal values, due to the high reliability (i.e. low failure probability) of the available transport technologies.

5.3. Sensitivity analysis

The sensitivity of the resilient supply chain design to modeling assumptions is tested via two approaches. First, a global sensitivity analysis is carried out by varying, in the range [10%, 200%] of their reference values, the parameters quantifying: (i) the unit investment and operation costs, (ii) the carbon footprint (direct and indirect CO_2 emissions), (iii) the failure rate of pipeline and truck (i.e. the most relevant transport technologies), and (iv) the price and the carbon intensity of the electricity imported from the grid. The results of this global sensitivity analysis are reported in Fig. 5, which shows the overall sensitivity ranges (with black whiskers). Secondly, the sensitivity of the infrastructure-resilient 2030 CCTS supply chain to the unit costs of

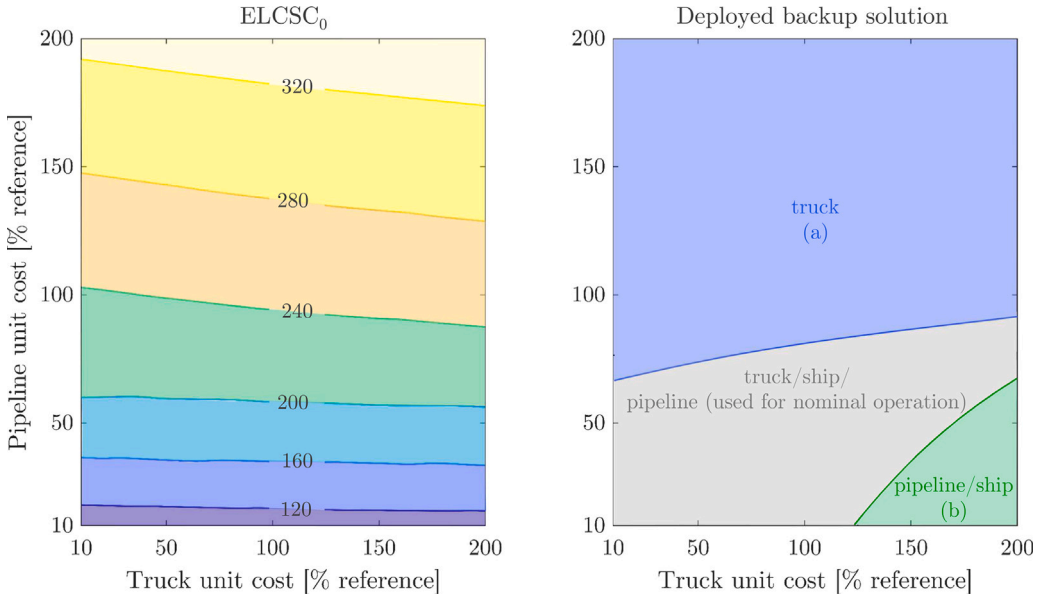


Fig. 7. Sensitivity of the infrastructure-resilient 2030 CCTS supply chain: ELCSC$_0$ and deployed backup solution as functions of the unit costs of trucks and pipelines. (a) and (b) refer to the supply chain configurations shown in Fig. 6.

trucks and pipelines, which are found to be the most relevant sensitivity parameters, is assessed; results are shown in Fig. 7, which shows the ELCSC$_0$ and the deployed backup solution of the resilient supply chain as functions of the track and pipeline unit costs.

The cost of resilient CCTS supply chains can vary significantly, as indicated by the whiskers in Fig. 5. In contrast, the CO_2 emissions of the CCTS supply chain are not subject to significant variations because of the low relevance of the emissions due to the deployment of the supply chain compared to the emissions of the WtE sector. Notably, a large cost increase is observed for the environment-resilient CCTS supply chain, which minimizes expected CO_2 emissions in 2030 (Fig. 5). This is because this configuration is entirely based on pipeline transport, both for cost-optimal and for backup connections, and implies the installation of long pipelines (from Basel to the North Sea storage site), which makes the design extremely sensitive to changes in pipeline cost. A lower impact is observed for 2040 and 2050, when the Swiss storage site becomes available.

Let us now further analyze the effects of solely varying the truck and pipeline unit costs. The shift upwards of the quasi-horizontal cost isolines shown on the left-hand panel of Fig. 7 indicates that the pipeline unit cost is the most relevant parameter to determine the LCSC of the supply chain, whereas the truck unit cost has a smaller impact. This is because infrastructure-resilient CCTS supply chains rely on pipelines under nominal operating conditions, and on trucks only during failures.

Let us now consider the right-hand panel of Fig. 7, which presents the choice of backup solutions deployed in the infrastructure-resilient 2030 CCTS supply chain for variable truck and pipeline unit costs. Three regions appear, which represent backup solutions deployed via trucks (design a in Fig. 6), trucks/ships (and pipelines that are also utilized in nominal operating conditions) and pipelines/ships (design b in Fig. 6). More specifically, whereas trucks are the only backup connections in reference conditions (Fig. 6a), an additional pipeline is installed, if its unit costs are small, in order to increase resilience against global contingency. In fact, once installed, such an additional pipeline is used for nominal operation due to its low operating costs, and trucks and ships (with the latter being originally installed for nominal operation) are used as backup connections (see Basel–Rotterdam–North Sea connections in Fig. 6b). When low pipeline costs are combined with high truck costs (e.g. to account for the cost of keeping a large fleet of trucks parked for long periods of time), backup pipelines are installed, and a meshed pipeline network is created, to be resilient also against local contingencies (Fig. 6b).

Similar considerations can be made for both CO_2 emissions and environment-resilient supply chains. However, in this case, changes in the chains configuration do not result in remarkable changes in CO_2 emissions, due to the small contribution of the chain components to the overall CO_2 emissions.

6. Conclusions

This study explores the design of resilient carbon capture, transport and storage (CCTS) supply chains aimed at decarbonizing industrial emitters, and focuses on the resilience of the CO_2 transport network. The resilience of a CCTS supply chain is defined as its ability to mitigate the effect of failures on its function to successfully capture, transport and permanently store CO_2 during a time horizon of interest, and its quantification relies on the concept of expected carbon not stored (ECNS).

An MILP optimization model is used to determine the optimal location and size of carbon capture units and the optimal structure of the CO_2 transport network that complies with a specified level of resilience, while minimizing the cost (infrastructure resilience) or the CO_2 emissions (environment resilience) of the supply chain. Whereas the model presented enables the design of resilient CCTS supply chains independently of their cost-optimal structure and can consider any failure scenario, we constrain the optimization algorithm to start from the minimum-cost solution and to consider one single failure at a time. This results in the (N-1)-resilient design of CCTS supply chains, i.e. a design that makes the cost-optimal solution fully resilient (it always stores all the CO_2 captured at the plants considered) against any single-connection failure.

While being general, the proposed optimization and analysis framework is presented and demonstrated by referring to a specific objective,

i.e., the decarbonization of the Swiss WtE sector from 2025 to 2050. The trade-off between minimum cost and maximum resilience, and the sensitivity of resilient CCTS supply chains on cost, emissions and reliability parameters, are investigated. Findings show that:

1. A multi-modal CO_2 network confers an overall advantage to the CCTS supply chain, as each transport technology performs best depending on the amount of CO_2 transported, on the transport distance, and on the type of use (i.e. nominal or backup connection). Combined, the properties of multiple transport modes can be leveraged to design a resilient network at different scales. More specifically, the low investment costs, high speed of transport and wide availability of trucks make them the best backup technology, allowing for a drastic reduction in the cost of resilience compared to the other modes of transport.
2. While a pipeline-only network proves to be the cost-optimal solution to decarbonize the Swiss WtE sector, a secondary truck network stands out as the most cost-effective contingency solution; furthermore, installing backup connections to support the longest and largest volume connections offers the best improvement in resilience.
3. Overall, (N-1)-resilience is achieved with a cost increase with respect to the cost-optimal solution ranging from about 5% (infrastructure resilience, based on backup truck connections) to about 70% (environment resilience, based on backup pipeline connections, hence in a meshed pipeline network) when the storage site is in the North Sea (i.e. in 2030); when the storage site is in Switzerland (i.e. in 2040 and 2050), the cost increase is much reduced and ranges from about 1% (infrastructure resilience) to about 10% (environment resilience).
4. Such resilient solutions allow to avoid venting CO_2 into the atmosphere in case of failures, hence reducing the expected CO_2 emissions of the cost-optimal supply chain by 50–100 $ktCO_2$ per year in 2030 (about 1.5%–3% of the CO_2 emissions of the WtE sector in 2030) and 200–250 $ktCO_2$ per year in 2050 (about 25%–32% of the CO_2 emissions of the WtE sector in 2050).
5. Our findings prove to be relatively robust against variations in cost, emissions and reliability parameters, though changes in the structure of resilient supply chains are observed when varying the values of such parameters. For example, the choice of trucks as the optimal transport mode for backup transport might change when considering additional types of costs (for example, the costs of keeping a large fleet of trucks, 3000 here, parked and non-operational for long periods of time).

While not considering all possible strategies to improve resilience, this study highlights the importance of designing resilient CCTS supply chains and provides a quantitative assessment of the cost of resilience. Moreover, it shows that resilience becomes more relevant and cheaper when considering larger CO_2 networks, possibly at an international scale and shared among multiple stakeholders, and larger amounts of CO_2 transported. We are confident that the concepts, models, and methods presented in this paper will be of value in the design, analysis, and optimization of the shared infrastructure for the management of CO_2, which is much required to deal with the climate crisis and to address the need to capture, collect, transport, and store CO_2 in an effective and resilient manner.

The analysis presented in this work can be expanded in multiple directions. First, the resilience of CO_2 capture and storage facilities can be included in the analysis of the overall resilience of CCTS supply chains. Moreover, a time resolution finer than one year can be investigated. This would allow to consider aspects such as temporary CO_2 storage in buffer tanks, time-dependent fluctuations of pipeline flow rates, and lag times in deploying the backup connections. Finally, the relationship between the existing and the optimal infrastructures can be studied to assess the potential of repurposing of existing infrastructure for carbon management.

CRediT authorship contribution statement

Paolo Gabrielli: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition. **Jordi Campos:** Formal analysis, Software, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Viola Becattini:** Methodology, Visualization, Writing – review & editing. **Marco Mazzotti:** Methodology, Visualization, Writing – review & editing. **Giovanni Sansavini:** Methodology, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Values for resilience analysis

The input data to the optimization problem used to assess and maximize the resilience of CCTS value chains, specifically the recovery time, δ and the failure rate, λ, are reported in Table A.1. All other input data to the optimization problem are reported in the Appendix of Becattini et al. (2022). The results of the resilience assessment, specifically the values of nominal and expected LCSC, of nominal and expected emissions, E, and of ECNS shown in Fig. 5, are reported in Table B.3.

Table A.1
Numerical values of the input data used to assess the resilience of CCTS value chains (Zep, 2011; Jo and Crowl, 2008; Duncan and Wang, 2014; Expert interviews, 2021).

Technology	δ [days]	λ [1/(km year)]
Truck	2	1.0E−04
Train (CH)	1	1.0E−04
Train (EU)	1	1.0E−04
Barge	10	1.0E−05
Ship	10	1.0E−05
Pipeline	20	1.3E−04

Appendix B. Emissions of Swiss WtE plants

The CO_2 emissions of all Swiss WtE plants in 2018 and corresponding hubs in the simplified network of Section 5.1 are reported in Table B.4.

Table B.3
Results of the resilience assessment (values needed to reproduce Fig. 5).

	Cost						Emissions						Resilience
	$LCSC_{nom}$			ELCSC			E_{nom} [$MtCO_2$]			E [$MtCO_2$]			ECNS [$MtCO_2$]
	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	
Simple network													
Cost-optimal	–	137	–	–	–	–	–	0.54	–	–	–	–	0.22
Infrastructure	–	148	–	–	152	–	–	0.64	–	–	0.65	–	0 ($R_0 = 1$)
Environment	–	195	–	–	195	–	–	0.56	–	–	0.56	–	0 ($R_0 = 1$)
2030													
Cost-optimal	–	234	–	–	–	–	–	3.49	–	–	–	–	0.10
Infrastructure	100	243	340	103	245	344	3.49	3.54	3.58	3.49	3.54	3.58	0 ($R_0 = 1$)
Environment	106	390	607	107	392	607	3.48	3.48	3.50	3.48	3.49	3.50	0 ($R_0 = 1$)
2040													
Cost-optimal	–	121	–	–	–	–	–	1.93	–	–	–	–	0.16
Infrastructure	76	122	173	76	122	173	1.92	1.94	1.97	1.92	1.94	1.97	0 ($R_0 = 1$)
Environment	76	135	173	76	135	173	1.87	1.89	1.91	1.88	1.90	1.92	0 ($R_0 = 1$)
2050													
Cost-optimal	–	121	–	–	–	–	–	0.55	–	–	–	–	0.22
Infrastructure	76	122	172	76	122	172	0.54	0.57	0.60	0.54	0.57	0.60	0 ($R_0 = 1$)
Environment	77	135	176	77	135	176	0.51	0.52	0.54	0.51	0.52	0.54	0 ($R_0 = 1$)

Table B.4
CO_2 emissions of all Swiss WtE plants in 2018 (Verband der Betreiber Schweizerischer Abfallverwertungsanlagen (VBSA), 2019), and corresponding hubs in the simplified network of Section 5.1.

Location	CO_2 emissions [tCO_2/y]	Simple supply chain hub
Basel	257,125	Basel (257 $ktCO_2/y$)
Geneva	230,402	Geneva (230 $ktCO_2/y$)
Giubiasco	188,190	Giubiasco (188 $ktCO_2/y$)
Trimmis	124,446	Luzern (3.8 $MtCO_2/y$)
Colombier	64,740	
Zuchwil	252,071	
Weinfelden	163,825	
Winterthur	224,196	
Buchs (AG)	157,025	
Oftringen	80,390	
Turgi	134,263	
Bern	156,229	
Biel	55,004	
Thun	146,573	
Posieux	114,048	
Niederurnen	123,038	
La Chaux-de-Fonds	60,129	
Buchs (SG)	208,719	
St. Gallen	82,044	
Lausanne	209,154	
Gamsen	43,400	
Sion	65,319	
Monthey	164,588	
Dietikon	105,275	
Hinwil	216,715	
Horgen	41,910	
ZH Hagenholz	270,227	
ZH Josefstrasse	122,829	
Luzern	302,944	
Bazenheid	98,558	

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